Brief Intro to Hardware Attacks

HTB Academy Module Notes · Tier 0 · Mini-Module · Bluetooth, Cryptanalysis Side-Channels, Microprocessor Vulnerabilities

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These notes are built from the module text you shared — the module overview page, Section 1 (Introduction), and Section 2 (Introduction to Bluetooth) — paraphrased and reorganised into a study reference with diagrams. Sections 3 onward aren't included yet; send their content and this can be extended the same way.

Module Overview

This mini-module concisely introduces hardware attacks, covering Bluetooth risks and attacks, cryptanalysis side-channel attacks, and vulnerabilities like Spectre and Meltdown. It delves into both historical and modern Bluetooth hacking techniques, explores the principles of cryptanalysis and different side-channel attacks, and outlines microprocessor design, optimisation strategies and vulnerabilities such as Spectre and Meltdown.

Attribute	Value
Difficulty	Medium
Tier	Tier 0
Estimated time	4 hours
Sections	8
Includes	Badge of Completion, 10 Cubes
Creator	PandaSt0rm
Rating	5.0 (5 reviews)

#	Section	Status
1	Introduction	Covered below
2	Introduction to Bluetooth	Covered below
3	Bluetooth Legacy Attacks	Not yet provided
4	Modern Bluetooth Attacks and Mitigation	Not yet provided
5	Introduction to Cryptanalysis	Not yet provided
6	Cryptanalysis Side-Channel Attacks	Not yet provided
7	Understanding Microprocessors	Not yet provided
8	Microprocessor Vulnerabilities	Not yet provided

Section 1 — Introduction

The module frames hardware security as a distinct problem from software security: some vulnerabilities live in the physical technology itself — from short-range wireless like Bluetooth up to the microchips every device runs on — and no software update can fully resolve them. The module takes a theoretical focus across three areas: Bluetooth hacking methods, cryptanalysis side-channel attacks, and microprocessor vulnerabilities.

Bluetooth hacking — the techniques named

- **`Bluesnarfing`** — A cyber-attack involving unauthorised access to information from wireless devices through Bluetooth.
- **`Bluejacking`** — An attack that sends unsolicited messages to Bluetooth-enabled devices.
- **`BlueSmacking`** — A Denial-of-Service attack that overwhelms a device's Bluetooth connection.
- **`Bluebugging`** — A technique used to gain control over a device via Bluetooth.
- **`BlueBorne`** — A set of vulnerabilities that allow attackers to take control of devices, spread malware, or perform other malicious activities via Bluetooth.
- **`KNOB`** — Key Negotiation of Bluetooth — manipulates the data encryption process during Bluetooth connection establishment, weakening security.
- **`BIAS`** — Bluetooth Impersonation AttackS — exploits a vulnerability in the pairing process, allowing an attacker to impersonate a trusted device.

Cryptanalysis side-channel attacks — introduced

These attacks utilise information gained from implementing and running a computer system, rather than brute force or theoretical weaknesses in algorithms. The module covers a short history of side-channel attacks, then two specific types:

- **`Timing Attacks`** — Exploit the correlation between the computation time of cryptographic algorithms and the secrets they process.
- **`Power-Monitoring Attacks`** — Monitor the power consumption of a device to determine what data it is processing.

Microprocessor vulnerabilities — introduced

Microprocessors form the backbone of any computational device. Their complex design and optimisation strategies often introduce vulnerabilities. The module explores what microprocessors are, then two notorious examples — **Spectre** and **Meltdown** — along with mitigation strategies:

- **`Retpoline`** — A binary modification technique used to thwart branch target injection.
- **`Compiler modifications`** — Compiler-level changes that help prevent speculative-execution leaks.
- **`Kernel Page Table Isolation (KPTI)`** — A technique used to isolate the kernel's memory space from user-space processes.
- **`Microcode updates`** — CPU firmware updates that patch vulnerable speculative-execution behaviour.

Section 2 — Introduction to Bluetooth

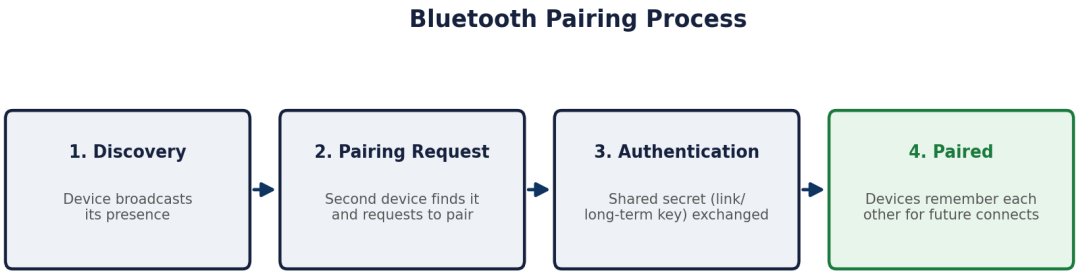
Bluetooth is a wireless technology standard designed for transferring data over short distances from fixed and mobile devices. It establishes personal area networks (PANs) using radio frequencies in the ISM band from 2.402 GHz to 2.480 GHz. Conceived as a wireless alternative to RS-232 data cables, it's named after the epithet of Harald Bluetooth, the tenth-century king who unified Denmark and Norway. Annual device shipments were projected to exceed 7 billion by 2026, according to the Bluetooth 2022 Market Update.

The pairing process

Establishing a Bluetooth connection starts with pairing — two devices discovering each other and establishing a connection:

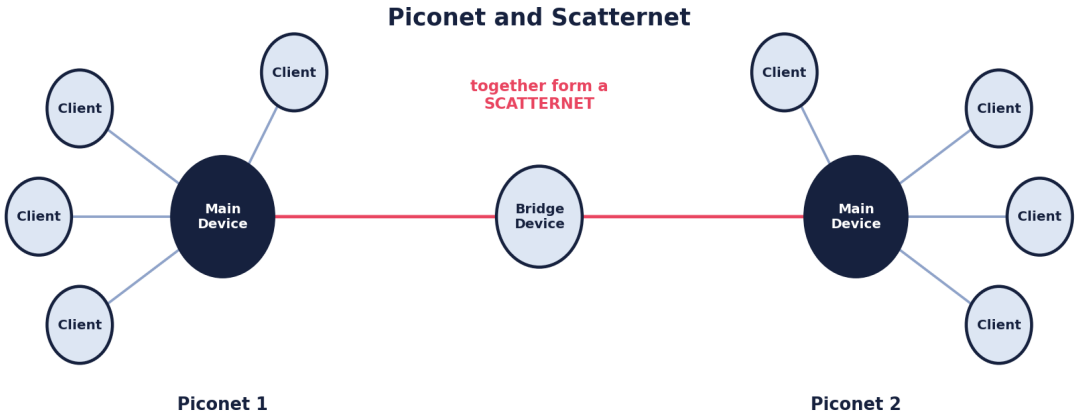
- **Discovery** — One device makes itself discoverable, broadcasting its presence to other Bluetooth devices within range.
- **Pairing Request** — A second device finds the discoverable device and sends a pairing request.
- **Authentication** — The devices authenticate each other through a shared secret — a link key or long-term key — which may involve entering a PIN on one or both devices.

Once paired, the devices remember each other's details and can automatically reconnect in future without repeating the pairing process.



Piconets and scatternets

After pairing, Bluetooth devices form a network known as a **piconet**: one main device and up to seven active client devices, with the main device coordinating communication and dictating the schedule for packet transmission. Multiple piconets can interact to form a larger network called a **scatternet**, where some devices act as bridges, participating in multiple piconets and enabling inter-piconet communication.



SCO vs ACL links

Bluetooth connections carry both data and audio, transmitted via packets on a schedule set by the piconet's main device. The specification defines two link types:

- **Synchronous Connection-Oriented (SCO)** — Used primarily for audio communication; reserves slots at regular intervals, guaranteeing steady, uninterrupted communication ideal for audio data.
- **Asynchronous Connection-Less (ACL)** — Used for all other types of data; does not reserve slots, instead transmitting whenever bandwidth allows.

Categories of Bluetooth risk

Risk, in a Bluetooth context, is any potential event that exploits the connection to compromise the confidentiality, integrity, or availability of data or systems — largely because the wireless nature of the technology allows remote, unauthorised access. Risks fall into several broad categories:

- **Unauthorised Access** — Unauthorised entities gain unsolicited access to Bluetooth-enabled devices, exploiting vulnerabilities to take control or eavesdrop on data exchanges.
- **Data Theft** — Attackers exploit vulnerabilities to extract stored or transmitted data without authorisation — contacts, messages, passwords, financial details, or other confidential data.
- **Interference** — Bluetooth shares the 2.4 GHz band with many other devices and technologies; malicious actors may disrupt or corrupt communication, causing data loss or connection instability.
- **Denial of Service (DoS)** — Overwhelming a device with excessive requests, or exploiting protocol vulnerabilities, to make the device unresponsive and unable to perform its functions.
- **Device Tracking** — Exploiting Bluetooth's radio signals to track a device's physical location, compromising owner privacy and potentially enabling stalking or other malicious activity.

Bluetooth attack reference table

A comprehensive overview of named Bluetooth attacks, as covered in this section:

Attack	Description
Bluejacking	A relatively harmless type of Bluetooth hacking where an attacker sends unsolicited messages or business cards to a device. Doesn't pose a direct threat, but can infringe on privacy and be a nuisance.

Bluesnarfing	Unauthorised access to a Bluetooth-enabled device's data, such as contacts, messages, or calendar entries, retrieved without the owner's knowledge or consent.
Bluebugging	Enables an attacker to control a Bluetooth device, including making calls, sending messages, and accessing data, by exploiting security weaknesses.
Car Whisperer	Targets vehicles specifically — attackers exploit Bluetooth vulnerabilities to remotely unlock car doors or even start the engine without physical access.
Bluesmacking & DoS	Leverages Bluetooth protocol vulnerabilities to disrupt or disable a connection. Bluesmacking specifically floods a target with excessive connection requests, overwhelming and disabling it.
Man-in-the-Middle	Intercepts and manipulates data exchanged between two Bluetooth devices by positioning the attacker between them, compromising confidentiality and integrity.
BlueBorne	Discovered in 2017 — a critical vulnerability allowing an attacker to take control of a device without requiring user interaction or pairing, exploiting multiple zero-days.
Key Extraction	Aims to retrieve the encryption keys used in a Bluetooth connection, letting an attacker decrypt and access sensitive data transmitted between devices.
Eavesdropping	Intercepting and listening to Bluetooth communications, capturing and analysing data such as passwords, financial details, or personal conversations.
Bluetooth Impersonation	An attacker impersonates a trusted Bluetooth device to gain unauthorised access or deceive the user into connecting to a malicious device.

Note: this table is the general-overview catalogue from Section 2. The module's later sections (3 and 4) split these into "Legacy" and "Modern" categories with more technical depth and mitigation guidance — send that content when you reach it and this reference can be extended to match.